

MongoDB Atlas & Mongoose Schemas Architecture

Subject: MongoDB & SQL Database Systems

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STUDYVAULT VERIFIED ACADEMIC NOTES & STUDY GUIDE

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1. MONGOOSE SCHEMA & MODEL DEFINITION

Define schemas with validation, defaults, virtuals, and pre-save hooks.

2. DOCUMENT POPULATION & AGGREGATION PIPELINES

- Use `.populate("category")` to resolve referenced ObjectId documents.
- Aggregation pipeline stages: `$match`, `$group`, `$sort`, `$project` for database analytics.